A-A-53239
29 September 1987

COMMERCIAL ITEM DESCRIPTION

GRID, RADIOGRAPHIC

The General Services Administration has authorized the use of this commercial item description in preference to interim Federal Specification GG-0G-00650A, dated 25 September 1970.

This commercial item description covers a straight wafer-type radiographic grid, complete with carrying case, for use with an x-ray apparatus during radiographic examination.

Salient characteristics:

The radiographic grid shall be a straight wafer-type, with carrying case, for use with an x-ray apparatus during radiographic examination. the grid shall be in the following sizes:

Size I - Suitable for 10 by 12 inch film with approximate overall dimensions of 11 by 13 inches.

Size II - Suitable for 14 by 17 inch film with approximate overall dimensions of 15 by 18 inches.

GRID

The grid shall have a 38 inch focal-film distance, so as to provide for a working range of 34 to 44 inches. Minimum transmission through the outer edges of the grid shall not be less than 70 percent of the transmission through the center of the grid. The grid ratio shall be 6:1. The grid shall have sixty lines to the inch.

The overall thickness of the completed grid shall be approximately 0.130 inch. Each corner of the completed grid shall be rounded.

The manufacturer's standard commercial finish shall be acceptable for the exterior cover sheets of the grid. The top cover sheet of the grid shall be suitably identified as "tube side" and shall have an inscribed center line running parallel to the long dimension.

AMSC N/A FSC 6525

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When tested, a radiograph of the grid shall show no marked variations in density, no opaque inclusions, and no pressure marks.

The grid cover shall be free of cracks, tears and deformed surfaces. The finish shall be uniform, free of scratches or other marks, and free of dirt, grease, or other foreign matter.

The surfaces of the grid shall be flat and free of warping. The grid shall be free of sharp edges and points. The grid marking shall be complete and legible.

Carrying Case.

The grid carrying case shall be designed to insure safe transportation and storage under field conditions. The case shall be designed to allow the grid to be easily removed, or inserted, with one hand. The case shall have secure, easy opening, latches or spring locks, to properly secure the grid. The inside of the case shall be lined with felt or foam sheet or strips, firmly cemented in place, to cushion and protect the grid.

There shall be no projections on the larger surfaces of the carrying case which interfere with stacking. The carrying case shall be free of warping and sharp edges or points.

The surface of the carrying case shall be free of indentations, burrs, cracks, or tears. The finish shall be uniform, free of scratches or other marks, and free of dirt, grease, or other foreign matter.

Any case marking shall be complete and legible.

Workmanship. The radiographic grid and carrying case shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One radiographic grid and carrying case, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one radiographic grid as specified in this description.

TABLE I. Sampling for examination.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total defects (Major and minor combined)	II	4.0
For dimensional examination	S-1	2.5

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Examination. The radiographic grid shall be examined to determine compliance with all requirements contained in this specification.

Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table II. Examination shall not be restricted to the classified possible defects below.

TABLE II. Classification of Defects.

Classification	Defect
Major A (Unit of product - Grid)	
101	Grid covers not free of cracks, tears and deformed surfaces.
102	Surfaces of grid not flat or not free of warping.
103	Grid not free of rattling which indicates that lead strips and dividers are not securely cemented between aluminum covers.
104	Grid not free of sharp edges or points.
105	Marking on grid not present, complete or legible.
Major B (Unit of product carrying case)	
151	Carrying case not capable of being opened and closed properly.
152	Grid not capable of being inserted or removed from case.
153	Foam or felt padding not securely cemented to inside of carrying case.
154	Latches or spring lock do not properly close and secure carrying case.
155	Latches or spring lock do not open on carrying case.
156	Carrying case not free of projections which interfere with stacking.
157	Carrying case not free of sharp edges or points.
158	Lid or case not free of warping.
Minor A (Unit of product-Grid)	
201	Grid not free of dirt, grease or other foreign matter.
202	Finish on grid not uniform or not free of scratches or marks.
203	Center line on grid not present, legible, or continuous.

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TABLE II. Classification of Defects - Continued.

Classification	Defect
Minor B	
(Unit of product-carrying case)	
251	Foam or felt padding not free of dirt, grease or other foreign matter.
252	Finish on carrying case not uniform or not free of scratches or marks.
253	Carrying case not free of dirt, grease or other foreign matter.
254	Surface of carrying case not free of indentations, burrs, cracks or tears.
255	Hinges not firmly secured to carrying case.

Dimensional Examination. The grids and carrying cases shall be examined for defects in dimensions. Any dimension not within the specifications herein shall be classified as a defect.

Tests.

Tests shall be conducted to determine compliance with all requirements contained in this specification.

Radiographic Test.

A radiograph shall be made at a target-to-film distance of 38 inches with central x-ray beam vertical to the plane of the grid and centered on the grid, and a maximum applied voltage of 100 KVP, the radiograph of the grid shall show no marked variations in density, no opaque inclusions, no lead turnover, no doubling nor omission of a lead strip, no pressure marks, nor sectional pattern. One radiograph shall be supplied with each grid and shall include the serial number of the grid.

Drop test. The carrying case with grid in place shall be subjected to the following test:

a. Four 18 inch drops to a concrete floor, once on each short edge of the carrying case, with the carrying case positioned like a diamond so that the line made up of two corners and one short edge as per figure will be parallel to the floor with the center of gravity directly over this line.

b. Two 36 inch drops to a concrete floor, with the case landing flat, once on each large side.

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Radiographic test shall be taken after drop test and compared with original radiograph.

After testing there shall be no damage to the grid; with the latches on carrying case remaining secure, and cemented felt or foam padding intact. Deformation of short edges shall not be cause for rejection provided the carrying case opens and closes properly. A radiographic test shall be taken after the drop test and compared with the original radiograph.

Metric products.

Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

6525-00-603-1250 (Size I)
6525-00-603-1310 (Size II)

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6525-1207

PGC 05157

Location: CID/GRID/H13-19

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53239	GRID, RADIOGRAPHIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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